[KING FAHAD CENTRAL HOSPITAL / STROKE CENTER LETTERHEAD]

Acute Stroke Complication Management Protocol

Standardized escalation protocol aligned with AHA/ASA Guidelines for the Early Management of Patients With Acute Ischemic Stroke, the 2022 AHA/ASA Spontaneous ICH Guideline, and American Stroke Association International Stroke Center Certification GL Stroke Center Certification requirements for complication tracking

RAPID RESPONSE / CODE
Activate for any Tier 1 event below
Patient / MRN: ______________
Event Date/Time: ______________

Document ID: SU-013-PROT-01 | Parent Policy: SU-013 — Critical Actions — Suspected Intracranial Hemorrhage / Systemic Bleeding After IV Thrombolytic Administration | Version: 1.0 | Controlled Copy

1 Purpose, Scope & Certification Alignment

This protocol standardizes recognition, immediate treatment, escalation, and documentation of complications occurring during or after acute stroke treatment (IV thrombolysis, mechanical thrombectomy, or general stroke unit care). It supports compliance with AHA/ASA Get With The Guidelines-Stroke adverse-event data elements and American Stroke Association International Stroke Center Certification GL stroke center certification requirements for a defined complication-management process with measurable time-to-intervention targets.

Applies to: Emergency Department, ICU / critical care/Stroke Unit, Neurointerventional Suite, and any unit administering IV alteplase/tenecteplase or performing endovascular thrombectomy.

2 Complication Escalation Tier Matrix

Tier	Triggering Findings	Required Response
Tier 1 — Life-Threatening	Symptomatic ICH; airway-threatening angioedema; malignant cerebral edema/herniation; major vascular perforation	Immediate rapid response / code activation; attending + neurosurgery + anesthesia at bedside
Tier 2 — Urgent	Access-site hematoma with hemodynamic change; new focal deficit suggesting reocclusion; significant GI/systemic bleeding	Stroke attending notified within 10 min; STAT imaging/labs; consult per protocol
Tier 3 — Non-Urgent Monitoring	Minor oral bleeding, mild access-site bruising, transient BP lability	Document, monitor per nursing protocol, notify team at next rounds

3 Symptomatic Intracranial Hemorrhage (sICH) After IV Thrombolysis

Tier 1 — activate rapid response immediately

Recognition / Diagnostic Criteria	Immediate Actions (STAT)
• Acute neurologic decline: ≥4-point rise in NIHSS from baseline or best post-treatment score • New headache, nausea/vomiting, acute hypertension, decreased level of consciousness • Any decline temporally associated with thrombolytic infusion or within 24–36 hours of dosing	**1.** STOP alteplase/tenecteplase infusion immediately if still running **2.** Emergent non-contrast head CT (do not delay for labs) **3.** Draw STAT: CBC/platelets, PT/INR, aPTT, fibrinogen, type & crossmatch **4.** Reassess NIHSS and vital signs; place on continuous cardiac/neuro monitoring **5.** Notify stroke attending, neurosurgery, and hematology/blood bank simultaneously

Medications & Reversal / Treatment Dosing

- Cryoprecipitate 10 units IV — repeat once if fibrinogen remains < 150 mg/dL
- Tranexamic acid 1000 mg IV over 10 min OR ε-aminocaproic acid 4–5 g IV, per institutional hematology protocol
- Platelet transfusion (1 unit apheresis) only if thrombocytopenic or on antiplatelet therapy per hematology guidance
- Treat hypertension to institutional post-ICH BP target; avoid hypotension
- Reverse any concurrent anticoagulant per agent-specific protocol (e.g., PCC/vitamin K for warfarin, idarucizumab for dabigatran, andexanet alfa where available for factor Xa inhibitors)

Escalation / Consults	Reassessment & Monitoring
• Neurosurgery — emergent evaluation for hematoma evacuation or ICP monitoring • Neurocritical care/ICU transfer • Repeat CT per neurosurgery/ICU direction (typically 6 and 24 hours) • Report event to stroke program medical director and quality/registry team within 24 hours	• Neuro checks and vitals every 15 min until stable, then per ICU protocol • Serial fibrinogen/coagulation panel until corrected and stable • Continuous ICP/neuro monitoring if hematoma evacuation performed

4 Orolingual Angioedema After IV Thrombolysis

Tier 1 if airway compromise — activate rapid response

Recognition / Diagnostic Criteria	Immediate Actions (STAT)
• Lip, tongue, or facial swelling — often contralateral to infarct hemisphere • Typically develops during infusion or within 1–2 hours; higher risk with ACE-inhibitor use • Assess for stridor, voice change, drooling, or oxygen desaturation (airway emergency)	**1.** Assess and secure airway first — anesthesia/ENT at bedside if any stridor or rapidly progressive swelling **2.** STOP alteplase/tenecteplase infusion if still running **3.** Elevate head of bed; apply supplemental oxygen **4.** Hold ACE inhibitors

Medications & Reversal / Treatment Dosing
• Methylprednisolone 125 mg IV • Diphenhydramine 50 mg IV • Famotidine 20 mg IV (H2 blocker) • Epinephrine 0.3 mg IM (1:1000) for rapidly progressive swelling or any airway compromise • Icatibant 30 mg subcutaneous if available and swelling refractory to above (bradykinin-mediated angioedema) • Fresh frozen plasma may be considered by specialist if refractory — contains C1 esterase inhibitor and ACE

Escalation / Consults	Reassessment & Monitoring
• Anesthesia/ENT for airway management or early intubation if progressive • Consider awake fiberoptic intubation over emergent cricothyrotomy planning if airway worsens • Notify stroke attending and document per adverse-event protocol	• Continuous airway/respiratory monitoring for minimum 24 hours or until swelling resolves • Repeat focused airway exam at 30-minute intervals during active swelling

5 Systemic (Extracranial) Major Bleeding

Tier 1–2 depending on hemodynamic stability

Recognition / Diagnostic Criteria	Immediate Actions (STAT)
• Overt bleeding (GI, GU, retroperitoneal, gingival, IV-site) with hemoglobin drop or hemodynamic change • Tachycardia, hypotension, or altered mental status without another explanation • Falling hemoglobin/hematocrit on serial labs	**1.** STOP thrombolytic infusion if still running; hold antiplatelet/anticoagulant doses **2.** Apply direct pressure to any accessible bleeding site **3.** Establish two large-bore IVs; STAT CBC, coagulation panel, fibrinogen, type & crossmatch **4.** IV fluid resuscitation per hemodynamic status

Medications & Reversal / Treatment Dosing
• Cryoprecipitate and/or antifibrinolytic (tranexamic acid) per hematology, as for sICH protocol • Packed red blood cell transfusion per institutional massive-transfusion/anemia thresholds • Reverse concurrent anticoagulants per agent-specific reversal protocol

Escalation / Consults	Reassessment & Monitoring
• General surgery / GI / interventional radiology consult based on bleeding source • Hematology/blood bank for transfusion support • ICU transfer if hemodynamically unstable	• Serial hemoglobin/hematocrit every 4–6 hours until stable • Continuous hemodynamic monitoring

6 Malignant Cerebral Edema / Herniation (Large-Vessel/Hemispheric Infarct)

Tier 1 — activate rapid response

Recognition / Diagnostic Criteria	Immediate Actions (STAT)
• Deteriorating level of consciousness, new pupillary asymmetry, or Cushing triad in large MCA/hemispheric infarct, typically 24–96 hours post-stroke • Radiographic midline shift or effacement of basal cisterns on follow-up CT	**1.** Elevate head of bed to 30°; keep head midline **2.** STAT non-contrast CT head **3.** Neurosurgery and neurocritical care notified immediately **4.** Avoid hypotonic fluids; correct hyponatremia if present

Medications & Reversal / Treatment Dosing
• Hyperosmolar therapy: mannitol 0.5–1 g/kg IV or hypertonic saline (e.g., 23.4% 30 mL or per institutional concentration) for acute herniation signs • Brief hyperventilation (target $PaCO_2$ 30–35 mmHg) only as temporizing bridge to definitive intervention

Escalation / Consults	Reassessment & Monitoring
• Emergent neurosurgical evaluation for decompressive hemicraniectomy, particularly in patients meeting age/timing criteria supported by AHA/ASA guidance • ICU transfer with ICP monitoring capability	• Neuro checks every 15–30 min • Repeat imaging per neurosurgery direction

7 Endovascular / Access-Site Complications (Post-Thrombectomy)

Tier 1–2 depending on severity

Recognition / Diagnostic Criteria	Immediate Actions (STAT)
• Groin/access-site hematoma, expanding swelling, new limb ischemia, or hypotension suggesting retroperitoneal bleed • Vessel dissection, perforation, or distal embolization noted on angiography • New neurologic decline suggesting reperfusion hemorrhage or reocclusion	**1.** Manual/direct pressure at access site; assess distal pulses **2.** STAT CBC, coagulation panel, type & crossmatch **3.** STAT CT (head ± abdomen/pelvis if retroperitoneal bleed suspected) **4.** Notify neurointerventional/vascular team immediately

Medications & Reversal / Treatment Dosing
• Reversal of periprocedural anticoagulation (e.g., protamine for heparin) per interventional team direction • Blood product support as needed per hematology

Escalation / Consults	Reassessment & Monitoring
• Interventional radiology/vascular surgery for access-site complications • Neurointervention team for in-procedure vessel injury or reocclusion • ICU transfer for hemodynamic instability	• Serial distal pulse and access-site checks per post-procedure protocol • Serial hemoglobin if retroperitoneal bleed suspected

8 General Supportive Complications (Aspiration, DVT/PE, Seizure)

Tier 2–3

Recognition / Diagnostic Criteria	Immediate Actions (STAT)
• Failed bedside dysphagia screen, witnessed aspiration, or new hypoxia/fever suggesting aspiration pneumonia • Unilateral limb swelling (DVT) or acute dyspnea/hypoxia/tachycardia (possible PE) • New convulsive activity or unexplained fluctuating consciousness (possible seizure)	**1.** Keep NPO and reassess airway/oxygenation for suspected aspiration **2.** STAT vitals, oxygen saturation, and physician evaluation **3.** For suspected seizure: protect airway, place patient in lateral position, time the event

Medications & Reversal / Treatment Dosing
• Formal speech-language pathology dysphagia evaluation prior to resuming oral intake • Empiric anticoagulation or IVC filter evaluation per institutional VTE protocol if thrombolysis contraindicated • Benzodiazepine (e.g., lorazepam 4 mg IV) for active seizure per institutional status epilepticus protocol; STAT EEG if indicated

Escalation / Consults	Reassessment & Monitoring
• Pulmonology/critical care for respiratory decline • Neurology for suspected seizure or non-convulsive status • Vascular medicine for confirmed DVT/PE	• Daily dysphagia and mobility/VTE risk reassessment throughout admission • Continuous pulse oximetry for at-risk patients

9 Documentation, Registry Reporting & Quality Review

- Document complication type, recognition time, intervention times, and outcome in the medical record and stroke registry (Get With The Guidelines-Stroke) within the encounter.
- All Tier 1 events require completion of an internal adverse-event/variance report and review at the next stroke program performance improvement (PI) committee meeting, per certification requirements.
- Time-to-recognition and time-to-intervention for each complication should be abstracted and trended for the stroke center's quarterly quality dashboard.
- Any sICH, procedural complication, or death associated with stroke treatment must be reviewed for root-cause analysis and reported per hospital patient-safety policy and applicable state/registry reporting requirements.

10 Protocol Approval & Review

Approved By (Stroke Medical Director)	Signature	Date
Pharmacy & Therapeutics Review	**Nursing / PI Committee Review**	**Next Scheduled Review Date**

Form Reference: Drug doses and thresholds above reflect commonly cited institutional protocols consistent with AHA/ASA Guidelines for the Early Management of Patients With Acute Ischemic Stroke and the 2022 AHA/ASA Guideline for the Management of Spontaneous Intracerebral Hemorrhage. This document is an administrative/clinical documentation template only — every medication, dose, and threshold must be reviewed and approved by the institution's stroke medical director, pharmacy & therapeutics committee, and neurocritical care leadership, and reconciled against the current AHA/ASA guideline version, before use in patient care.